

Qns	Answer	Marks
4a	Electric force per unit positive charge on a small stationary test charge at that point.	B1
4bi	Considering the distance between the nearest two equipotential lines around point A (4V and 0V) $d = 3.5 \text{ cm}$ $E = \Delta V/d = (4-0)/(3.5 \times 10^{-2})$ $F = eE = 1.6 \times 10^{-19} \times (4-0)/(3.5 \times 10^{-2})$ $= 1.8 \times 10^{-17} \text{ N}$	M1 A1
4bii	From point B to C, $\Delta V = -6 - 4 = -10\text{V}$ Change in $E_P = e\Delta V = -1.6 \times 10^{-19} (-10)$ $= 1.6 \times 10^{-18} \text{ J (Gain)}$ By Principle of Conservation of Energy, Loss in Kinetic energy = Gain in Electric Potential Energy $\frac{1}{2} mu^2 - \frac{1}{2} mv^2 = 9.6 \times 10^{-19}$ $\frac{1}{2} (9.11 \times 10^{-31})(5.3 \times 10^6)^2 - \frac{1}{2} (9.11 \times 10^{-31})v^2 = 1.6 \times 10^{-18}$ $v = 5.0 \times 10^6 \text{ m s}^{-1}$	C1 C1 A1
c	The change in potential from P to Q can be determined by estimating the area under the graph, 1 square : $(0.5 \times 10^3) \text{ V m}^{-1} \times 10^{-2} \text{ m} = 5 \text{ V}$ 18 squares : $18 \times 5 = 90 \text{ V}$ Change in potential energy = $1.6 \times 10^{-19} \times 90 = 1.44 \times 10^{-7} \text{ J}$	C1 C1 A1

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5(a)	1/0.50	